

[Home](#) > [Proceedings of International Conference on Artificial Intelligence, Communication Technologies and Smart Cities](#) > [Conference paper](#)

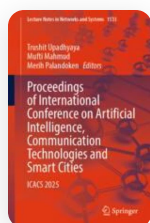
Detecting Bangla DeepFake Audio: A Dual Approach Using LSTM and WaveNet

| [Conference paper](#) | First Online: 28 October 2025

| pp 571–588 | [Cite this conference paper](#)

 [Save conference paper](#)

[View saved research](#) >



[Proceedings of International
Conference on Artificial
Intelligence, Communication...](#)
(ICACS 2025)

[Nafis Sadique Ayan](#) , [Md. Akteruzzaman Dipto](#), [Sultana Razia Faria](#), [Sumaiya Akhtar Mitu](#) & [Shah Murtaza Rashid Al Masud](#)

 Part of the book series: [Lecture Notes in Networks and Systems](#) ((LNNS, volume 1533))

 Included in the following conference series:
[International Conference on Artificial Intelligence, Communication Technologies & Smart Cities](#)

 329 Accesses

 The original version of the chapter has been revised. A correction to this chapter can be found at https://doi.org/10.1007/978-981-96-9196-8_46

Abstract

In the age of Artificial Intelligence, the rise of AI-generated audios, also known as audio deepfakes, causes significant concerns, especially regarding the authenticity of voices. While some research has been conducted on AD audio deepfake detection, the focus has mainly been on languages like English, leaving a gap in the study of Bangla-specific deepfake detection. This paper pioneers the exploration of audio deepfake detection in Bangla by developing two models: a Recurrent Neural Network (RNN)-based Long Short-Term Memory (LSTM) using MFCC features and a custom WaveNet model using normalized raw audio files. The LSTM model achieved an accuracy of 89%, while WaveNet outperformed it with a 91% accuracy rate. This research faced several unique challenges. In the absence of a proper deepfake audio dataset, a custom dataset was made for this research work. Additionally, the phonetic and tonal complexity of the Bangla language posed difficulties in distinguishing subtle variations between authentic and synthetic speech. The diverse accents and dialects within Bangla further complicated model generalization, requiring robust preprocessing and feature extraction techniques. Despite these challenges, this research successfully identified differences between authentic and synthetic audio, significantly improving detection accuracy for the Bangla language. This work not only advances audio deepfake detection in Bangla but also underscores the need for linguistic diversity in technological innovation, paving the way for future studies in other under-represented languages.

 This is a preview of subscription content, [log in via an institution](#)  to check access.

Access this chapter

Log in via an institution

Subscribe and save

✔

Springer+

from €37.37 /Month

Starting from 10 chapters or articles per month
Access and download chapters and articles from more than 300k books and 2,500 journals
Cancel anytime

View plans →

Buy Now

^

Chapter

EUR 29.95

Price includes VAT (Bangladesh)

Available as PDF
Read on any device
Instant download
Own it forever

Buy Chapter

▼ eBook

EUR 213.99

▼ Softcover Book

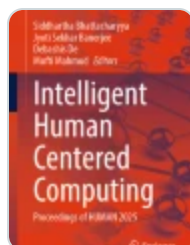
EUR 249.99

Tax calculation will be finalised at checkout

Purchases are for personal use only

[Institutional subscriptions](#) →

Similar content being viewed by others



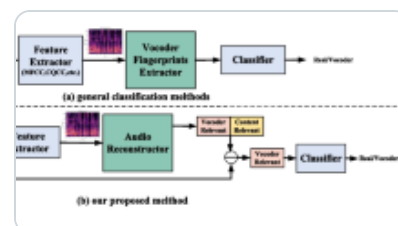
Detecting Synthetic Voices: A Novel Approach to Audio Deepfake Identification

Chapter | © 2026



A Comparative Analysis of Advanced Deep Learning Techniques for Detecting Deepfake...

Chapter | © 2025



Understanding and leveraging vocoder fingerprints for synthetic speech...

Article | 21 April 2025

Explore related subjects

Discover the latest articles, books and news in related subjects, suggested using machine learning.

[African Languages](#)

[Asian Languages](#)

[Audio Journalism](#)

[Minority Languages](#)

[Speech and Audio Processing](#)

[Speech and Audio Signal Processing](#)

[Speaker Recognition Systems and Techniques](#)

Change history

• 17 April 2026

A correction has been published.

References

1. Shaaban OA, Yildirim R, Alguttar AA (2023) Audio deepfake approaches. IEEE Access 11:132652–132682. <https://doi.org/10.1109/ACCESS.2023.3333866>
2. Almutairi Z, Elgibreen H (2022) A review of modern audio deepfake detection methods: challenges and future directions. Algorithms 15(5):155
3. Mateen M, Bawany N (2023) Deep learning approach for detecting audio deepfakes in Urdu. NUML Inte J Eng Comput 2:1–11. <https://doi.org/10.52015/nijec.v2i1.37>
4. Ballesteros DM, Rodriguez-Ortega Y, Renza D, Arce G (2021) Deep4snet: deep learning for fake speech classification. Expert Syst Appl 184:115465
5. Hamza A, Javed ARR, Iqbal F, Kryvinska N, Almadhor AS, Jalil Z, Borghol R (2022) Deepfake audio detection via MFCC features using machine learning. IEEE Access 10:134018–134028
6. Liu T, Yan D, Wang R, Yan N, Chen G (2021) Identification of fake stereo audio using SVM and CNN. Information 12(7):263
7. Almutairi Z, Elgibreen H (2023) Detecting fake audio of Arabic speakers using self-supervised deep learning. IEEE Access 11:72134–72147

8. Rodríguez-Ortega Y, Ballesteros DM, Renza D (2020) A machine learning model to detect fake voice. In: International conference on applied informatics. Springer, Berlin, pp 3–13
9. Wang C, Yi J, Tao J, Sun H, Chen X, Tian Z, Ma H, Fan C, Fu R (2022) Fully automated end-to-end fake audio detection. In: Proceedings of the 1st international workshop on deepfake detection for audio multimedia, pp 27–33
10. Nguyen QT, Bui TD (2016) Speech classification using sift features on spectrogram images. Vietnam J Comput Sci 3:247–257
11. Hua G, Teoh ABJ, Zhang H (2021) Towards end-to-end synthetic speech detection. IEEE Signal Process Lett 28:1265–1269
12. Lataifeh M, Elnagar A, Shahin I, Nassif AB (2020) Arabic audio clips: identification and discrimination of authentic cantillations from imitations. Neurocomputing 418:162–177
13. Yi J, Fu R, Tao J, Nie S, Ma H, Wang C, Wang T, Tian Z, Bai Y, Fan C, et al (2022) Add 2022: the first audio deep synthesis detection challenge. In: ICASSP 2022–2022 IEEE international conference on acoustics, speech and signal processing (ICASSP). IEEE, USA, pp 9216–9220
14. Sahidullah M, Kinnunen T, Hanilçi C (2015) A comparison of features for synthetic speech detection. In: 16th Annual conference of the international speech communication association (INTERSPEECH 2015), pp 2087–2091

15. Wu Z, Chng ES, Li H (2012) Detecting converted speech and natural speech for anti-spoofing attack in speaker recognition. In: Thirteenth annual conference of the international speech communication association
16. Muda L, Begam M, Elamvazuthi I (2010) Voice recognition algorithms using MEL frequency cepstral coefficient (MFCC) and dynamic time warping (DTW) techniques. arXiv preprint [arXiv:1003.4083](https://arxiv.org/abs/1003.4083)
17. Balamurali B, Lin KE, Lui S, Chen JM, Herremans D (2019) Toward robust audio spoofing detection: a detailed comparison of traditional and learned features. IEEE Access 7:84229–84241
18. Gao Y, Lian J, Raj B, Singh R (2021) Detection and evaluation of human and machine generated speech in spoofing attacks on automatic speaker verification systems. In: 2021 IEEE spoken language technology workshop (SLT). IEEE, USA, pp 544–551
19. Chen T, Kumar A, Nagarsheth P, Sivaraman G, Khoury E (2020) Generalization of audio deepfake detection. Odyssey 132–137
20. Dixit A, Kaur N, Kingra S (2023) Review of audio deepfake detection techniques: Issues and prospects. Expert Syst e13322
21. Repository OS (2023) Coqui-ai. <https://github.com/coqui-ai/TTS.git>. Accessed: 05 Nov 2023
22. Hossen M (2023) Comprehensive bangla TTs. <https://github.com/mobassir94/comprehensive-bangla-tts.git>. Accessed: 05 Nov 2023

Acknowledgements

We extend our heartfelt gratitude to the Department of Computer Science and Engineering, University of Asia Pacific, and all those who supported us in shaping this work.

Author information

Authors and Affiliations

Department of Computer Science and Engineering, University of Asia Pacific, Dhaka, Bangladesh

Nafis Sadique Ayan, Md. Akteruzzaman Dipto, Sultana Razia Faria, Sumaiya Akhtar Mitu & Shah Murtaza Rashid Al Masud

Corresponding author

Correspondence to [Nafis Sadique Ayan](#).

Editor information

Editors and Affiliations

Electronics & Communication Engineering, CHARUSAT, Changa, Gujarat, India
Trushit Upadhyaya

Nottingham Trent University, Nottingham, UK
Mufti Mahmud

Department of Electrical & Electronic Engineering, Izmir Katip Celebi University, Izmir, Türkiye
Merih Palandoken

Ethics declarations

Disclosure of Interests

The authors have no competing interests to declare that are relevant to the content of this article.

Rights and permissions

[Reprints and permissions](#)

Copyright information

© 2026 The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd.

About this paper

Cite this paper

Ayan, N.S., Dipto, M.A., Faria, S.R., Mitu, S.A., Al Masud, S.M.R. (2026). Detecting Bangla DeepFake Audio: A Dual Approach Using LSTM and WaveNet. In: Upadhyaya, T., Mahmud, M., Palandoken, M. (eds) Proceedings of International Conference on Artificial Intelligence, Communication Technologies and Smart Cities. ICACS 2025. Lecture Notes in Networks and Systems, vol 1533. Springer, Singapore. https://doi.org/10.1007/978-981-96-9196-8_38

[.RIS](#) [.ENW](#) [.BIB](#)

DOI	Published	Publisher Name
https://doi.org/10.1007/978-981-96-9196-8_38	28 October 2025	Springer, Singapore

Print ISBN	Online ISBN	eBook Packages
978-981-96-9195-1	978-981-96-9196-8	Intelligent Technologies and Robotics
		Intelligent Technologies and Robotics (R0)

[Springer Nature](#)
[Proceedings excluding](#)
[Computer Science](#)

Keywords

[Artificial intelligence](#)

[Deep learning](#)

[Audio](#)

[Audio deepfake](#)

[Audio forensic](#)

[RNN](#)

[RNN based LSTM](#)

[WaveNet](#)

[MFCC features](#)

Publish with us

[Policies and ethics](#) 